

Section 5: Multiple Linear Regression

STAT 139 Teaching Team

1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See Canvas for the most accurate information!

1.3 Attendance

- Please notify the Teaching Fellow (TF) if your attendance hasn't been taken!

2 Multiple Linear Regression Model

Idea. We're now ready to extend to *multiple linear regression*, where we have p predictors instead of just one! It will be easier to write our model in terms of vectors and matrices.

2.1 Helpful Linear Algebra

- $\vec{x}^\top A \vec{y} = \vec{y}^\top A^\top \vec{x}$ for any matrix A and vectors $\vec{x}$ and $\vec{y}$ of compatible dimensions.¹
- $(cA)^\top = cA^\top$ for any matrix A and scalar c .
- $(ABC)^\top = C^\top B^\top A^\top$ for any matrices A, B , and C .
- $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ for any invertible matrices A, B , and C .
- $(A^\top)^{-1} = (A^{-1})^\top$ for any invertible matrix A .
- $A^\top = A$ for any symmetric matrix A .
- $\mathbb{E}[A\vec{X}] = A\mathbb{E}[\vec{X}]$ for any matrix A and random vector $\vec{X}$.
- $\text{Var}[A\vec{X}] = A\text{Var}[\vec{X}]A^\top$ for any matrix A and random vector $\vec{X}$.
- 💡: Geometrically, a matrix represents a linear transformation of space.
- ⚡: Matrix multiplication is associative—i.e., $ABC = (AB)C$ —but *not* commutative—i.e., $AB \neq BA$.

2.2 Fundamentals

Definition 1 (Model in matrix notation). Suppose that we observe n observations, each with p predictors. Then our *model* is

¹Unless otherwise noted, a “matrix” refers to a constant (i.e., non-random) matrix.

$$\vec{Y}_{n \times 1} = \mathbf{X}_{n \times (p+1)} \vec{\beta}_{(p+1) \times 1} + \vec{\varepsilon}_{n \times 1},$$

where

$$\vec{Y}_{n \times 1} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}$$

is the vector of *outcomes*,

$$\mathbf{X}_{n \times (p+1)} = \begin{pmatrix} 1 & x_{1,1} & \dots & x_{1,p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n,1} & \dots & x_{n,p} \end{pmatrix}$$

is the *design matrix* such that $x_{i,j}$ is predictor j for observation i ,

$$\vec{\beta}_{(p+1) \times 1} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}$$

is the vector of *coefficients*, and

$$\vec{\varepsilon}_{n \times 1} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

is the vector of *errors*.

Definition 2 (Model for individual observation). Equivalently, our *model* is

$$Y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j} + \varepsilon_i$$

for all $i \in \{1, \dots, n\}$.

•  There are p predictors and 1 intercept, so the column of 1s corresponds to β_0 . These are two equivalent ways to express the same model!

 **Concept Checker 1.** Assume linearity. In the multiple linear regression model, what is the interpretation of β_0 ? How about β_j for all $j \in \{1, \dots, p\}$?

Solution

We can rewrite our model as $\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}$. Thus, β_0 is the *expected* outcome when all predictors $(x_{i,1}, \dots, x_{i,p})$ equal 0. Next, β_j is the expected change in

Y_i associated with a one-unit increase in $x_{i,j}$ (i.e., predictor j), holding all the other predictors constant. We can show this by taking a partial derivative with respect to $x_{i,j}$.

3 Inference

Idea. Like with simple linear regression, we must *infer* the best estimates for the unknown coefficients from the data. Thankfully, we use the *same assumptions* as before.

3.1 Assumptions

Definition 3 (Existence). $\text{Var}[\varepsilon_i] < \infty$ for all $i \in \{1, \dots, n\}$.

Definition 4 (Linearity). $\mathbb{E}[Y_i] = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j}$, which implies that $\mathbb{E}[\varepsilon_i] = 0$ for all $i \in \{1, \dots, n\}$.

Definition 5 (Independence). $\varepsilon_i \perp\!\!\!\perp \varepsilon_j$ for all $i \neq j$.

Definition 6 (Homoskedasticity). $\text{Var}[\varepsilon_i] = \sigma^2$ for all $i \in \{1, \dots, n\}$.

Definition 7 (Normality). $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ for all $i \in \{1, \dots, n\}$.

Definition 8 (ELIHN). Together, the five main assumptions can be summarized as $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, with $\sigma^2 < \infty$.

3.2 Prediction

Definition 9 (Predicted outcome). Suppose that we have $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$, and $\hat{\sigma}^2$ as estimates from our data. Naturally, we calculate our *predicted outcome* as

$$\hat{Y}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{i,j}$$

for all $i \in \{1, \dots, n\}$.

- In matrix notation,

$$\hat{\vec{Y}} = \mathbf{X} \hat{\vec{\beta}}.$$

3.3 Ordinary Least-Squares (OLS) Estimation

Definition 10 (Objective function). What we want to *minimize* to form our estimators.

- Like before, we'll often use *sum of squares errors* (SSE).

Definition 11 (Sum of squares errors (SSE)).

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

- Since $\hat{Y}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{i,j}$, we can rewrite this as

$$\text{SSE} = \sum_{i=1}^n \left(Y_i - \left(\hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{i,j} \right) \right)^2.$$

- In matrix notation,

$$\text{SSE} = (\vec{Y} - \mathbf{X}\hat{\vec{\beta}})^\top (\vec{Y} - \mathbf{X}\hat{\vec{\beta}}).$$

 **Concept Checker 2.** Since $\hat{\vec{Y}} = \mathbf{X}\hat{\vec{\beta}}$, we can rewrite SSE as $(\vec{Y} - \hat{\vec{Y}})^\top (\vec{Y} - \hat{\vec{Y}})$. Show that $(\vec{Y} - \hat{\vec{Y}})^\top (\vec{Y} - \hat{\vec{Y}}) = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$.

Solution

$$\begin{aligned} \text{First, } \vec{Y} - \hat{\vec{Y}} &= \begin{pmatrix} Y_1 - \hat{Y}_1 \\ \vdots \\ Y_n - \hat{Y}_n \end{pmatrix}. \quad \text{Thus, } (\vec{Y} - \hat{\vec{Y}})^\top (\vec{Y} - \hat{\vec{Y}}) = \\ (Y_1 - \hat{Y}_1 \dots Y_n - \hat{Y}_n) &\begin{pmatrix} Y_1 - \hat{Y}_1 \\ \vdots \\ Y_n - \hat{Y}_n \end{pmatrix} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2. \end{aligned}$$

Definition 12 (OLS estimators).

$$(\hat{\beta}_{0,\text{OLS}}, \hat{\beta}_{1,\text{OLS}}, \dots, \hat{\beta}_{p,\text{OLS}}) = \arg \min_{(\beta_0, \beta_1, \dots, \beta_p) \in \mathbb{R}^{p+1}} \left(\sum_{i=1}^n \left(Y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j x_{i,j} \right) \right)^2 \right)$$

- I.e., our estimators are the values for $\beta_0, \beta_1, \dots, \beta_p$ that *minimize* SSE:

$$\hat{\vec{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{Y}.$$

 **Concept Checker 3.** The equation requires multiplying by $(\mathbf{X}^\top \mathbf{X})^{-1}$. When might this fail, and what does that mean in practice?

Solution

Recall that a matrix A does not have an inverse (i.e., A^{-1} does not exist) if it is not a square matrix, if its determinant is zero, or if its rows and columns are linearly dependent. Because $\mathbf{X}^\top \mathbf{X}$ is square by construction, it is invertible exactly when it has full rank. This occurs exactly when $\mathbf{X}$ has full column rank—i.e., its columns are linearly independent. If the columns of $\mathbf{X}$ are linearly dependent (perfect collinearity), $\mathbf{X}^\top \mathbf{X}$ is singular. Related, if its columns are close to linearly dependent, $\mathbf{X}^\top \mathbf{X}$ may be ill-conditioned, making the calculation numerically unstable.

3.4 Simple vs. Multiple Linear Regression

Quantity	Simple Linear Regression	Multiple Linear Regression
Model	$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$	$\vec{Y} = \mathbf{X}\vec{\beta} + \vec{\varepsilon}$
SSE	$\sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$	$(\vec{Y} - \mathbf{X}\hat{\vec{\beta}})^\top (\vec{Y} - \mathbf{X}\hat{\vec{\beta}})$
Coefficient OLS Estimator	$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$	$\hat{\vec{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{Y}$
Variance OLS Estimator	$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 = \frac{\text{SSE}}{n-2}$	$\hat{\sigma}_{\text{OLS}}^2 = \frac{1}{n-(p+1)} (\vec{Y} - \mathbf{X}\hat{\vec{\beta}})^\top (\vec{Y} - \mathbf{X}\hat{\vec{\beta}}) = \frac{\text{SSE}}{n-(p+1)}$

4 Geometry of Least Squares

Idea. We can use linear algebra to better understand the *geometric interpretation* of least squares, starting from the ground up.

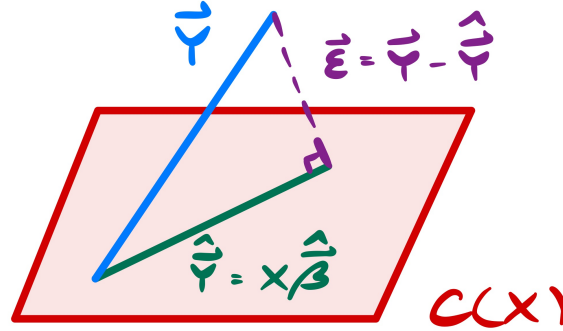


FIGURE 1: The linear projection of $\vec{Y}$ onto $C(\mathbf{X})$.

Definition 13 (Orthogonality). Two vectors $\vec{a}$ and $\vec{b}$ are *orthogonal* if they are perpendicular to each other. Equivalently,

$$\vec{a} \cdot \vec{b} = \vec{a}^\top \vec{b} = 0.$$

Definition 14 (Outcome and prediction vectors). Let $\vec{Y}$ be the vector of *outcomes* and let $\hat{\vec{Y}} = \mathbf{X}\hat{\vec{\beta}}$ be the vector of *predictions*.

- Minimizing SSE is equivalent to getting $\hat{\vec{Y}}$ as *close* as possible to $\vec{Y}$.

Definition 15 (Residual vector). Let $\vec{e} = \vec{Y} - \hat{\vec{Y}}$ be the vector of *residuals*.

- Minimizing SSE is equivalent to making $\vec{e}$ as *small* as possible.

Definition 16 (Column space). $C(\mathbf{X})$, the *column space* of $\mathbf{X}$, is the set of all possible linear combinations of its column vectors.

- As a linear combination, $\hat{\vec{Y}} = \mathbf{X}\hat{\vec{\beta}}$ must reside in $C(\mathbf{X})$.
- Minimizing SSE is equivalent to finding the $\hat{\vec{Y}} \in C(\mathbf{X})$ that is closest to $\vec{Y}$ in Euclidean distance—i.e., the ℓ_2 -norm of $\vec{e}$ is minimized.

Definition 17 (Linear projection). Geometrically, the best $\hat{\vec{Y}}$ is the *linear projection* of $\vec{Y}$ onto $C(\mathbf{X})$.

- I.e., $\vec{e}$ is orthogonal to every column of $\mathbf{X}$, which implies that $\mathbf{X}^\top \vec{e} = 0$.

Definition 18 (OLS estimators). We arrive at the same estimators:

$$\hat{\vec{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{Y}.^2$$

5 Adding the Normality Assumption

Idea. Like before, the fifth assumption of Normality allows us to extend our inference.

5.1 Point Estimation for Parameters

Definition 19 (ELIHN). Recall that the five main assumptions can be summarized as

$$\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

with $\sigma^2 < \infty$ for all $i \in \{1, \dots, n\}$. In matrix notation,

$$\vec{\varepsilon} \sim \text{MVN}(\vec{0}, \sigma^2 I_{n \times n}),$$

where $I_{n \times n}$ is the $n \times n$ identity matrix.

- Equivalently,

$$Y_i \stackrel{\text{ind.}}{\sim} \mathcal{N}\left(\beta_0 + \sum_{j=1}^p \beta_j x_{i,j}, \sigma^2\right).$$

In matrix notation,

$$\vec{Y} \sim \text{MVN}(\mathbf{X}\vec{\beta}, \sigma^2 I_{n \times n}).$$

 **Concept Checker 4.** What does $\sigma^2 I_{n \times n}$ look like as a covariance matrix? What is on the diagonal elements? What is on the off-diagonal elements?

²See Appendix A.1.

Solution

Visually, $\sigma^2 I_{n \times n} = \begin{pmatrix} \sigma^2 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma^2 \end{pmatrix}$, where each diagonal element is σ^2 while each off-diagonal element is 0. This makes sense since, by assumption, each error is independent (implying 0 covariance) and has the same variance.

5.2 t -Based Inference for Parameters

Definition 20 (Distributions). Under ELIHN, we have the following distributions for our estimators:

$$\begin{aligned}\widehat{\vec{\beta}} &\sim \text{MVN}\left(\vec{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right), \\ \hat{\beta}_j &\sim \mathcal{N}\left(\beta_j, \sigma^2(\mathbf{X}^\top \mathbf{X})_{(j+1, j+1)}^{-1}\right),\end{aligned}$$

and

$$\frac{n - (p + 1)}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-(p+1)}^2$$

for all $j \in \{0, \dots, p\}$.

Definition 21 (Variance-covariance matrix). From above,

$$\text{Var}[\widehat{\vec{\beta}}] = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1},$$

where $\sigma^2(\mathbf{X}^\top \mathbf{X})_{(j,k)}^{-1}$ corresponds to row j and column k in the matrix—i.e.,

$$\sigma^2(\mathbf{X}^\top \mathbf{X})_{(j,k)}^{-1} = \text{Cov}[\hat{\beta}_{j-1}, \hat{\beta}_{k-1}].$$

Definition 22 (Hypothesis test for coefficient). To test $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$, we use

$$T = \frac{\hat{\beta}_j - 0}{\hat{\sigma}_{\text{OLS}} \sqrt{(\mathbf{X}^\top \mathbf{X})_{(j+1, j+1)}^{-1}}} \stackrel{H_0}{\sim} t_{n-(p+1)}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

Definition 23 (Confidence interval for coefficient). The $100(1 - \alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{1-\frac{\alpha}{2}, n-(p+1)} \times \hat{\sigma}_{\text{OLS}} \sqrt{(\mathbf{X}^\top \mathbf{X})_{(j+1, j+1)}^{-1}}.$$

 **Concept Checker 5.** In multiple linear regression, $\hat{\beta}_j \sim \mathcal{N}\left(\beta_j, \sigma^2(\mathbf{X}^\top \mathbf{X})_{(j+1, j+1)}^{-1}\right)$. Show that this is equivalent to $\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right)$ in simple linear regression (where $p = 1$).

Solution

Let $p = 1$ and $j = 1$. By pattern-matching, we want to show that $\frac{\sigma^2}{s_{xx}} = \sigma^2(\mathbf{X}^\top \mathbf{X})_{(2,2)}^{-1} \iff \frac{1}{s_{xx}} = (\mathbf{X}^\top \mathbf{X})_{(2,2)}^{-1}$. First, $\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}$, so $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & \dots & 1 \\ x_1 & \dots & x_n \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}$. Since $\mathbf{X}^\top \mathbf{X}$ is a 2×2 matrix, $(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} \begin{pmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{pmatrix}$, so $(\mathbf{X}^\top \mathbf{X})_{(2,2)}^{-1} = \frac{1}{n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2} (n)$, which simplifies to $\frac{1}{\sum_{i=1}^n x_i^2 - \frac{1}{n}(\sum_{i=1}^n x_i)^2}$. Notice that $\sum_{i=1}^n x_i^2 - \frac{1}{n}(\sum_{i=1}^n x_i)^2 = \sum_{i=1}^n (x_i - \bar{x})^2$ by the sum of squares identity—i.e., $\sum_{i=1}^n (x_i - c)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - c)^2$ —with $c = 0$. Thus, $(\mathbf{X}^\top \mathbf{X})_{(2,2)}^{-1} = \frac{1}{s_{xx}}$.

5.3 Simple vs. Multiple Linear Regression

Quantity	Simple Linear Regression	Multiple Linear Regression
Coefficient OLS Estimator's Distribution	Slope: $\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{s_{xx}}\right)$, Intercept: $\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}\right)\right)$	Joint: $\hat{\vec{\beta}} \sim \text{MVN}\left(\vec{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right)$, Marginal: $\hat{\beta}_j \sim \mathcal{N}\left(\beta_j, \sigma^2(\mathbf{X}^\top \mathbf{X})_{(j+1,j+1)}^{-1}\right)$
Variance OLS Estimator's Distribution	$\frac{n-2}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-2}^2$	$\frac{n-(p+1)}{\sigma^2} \hat{\sigma}_{\text{OLS}}^2 \sim \chi_{n-(p+1)}^2$
Confidence Interval for Mean Response at x_0 or $\vec{x}_0 = (1, x_{0,1}, \dots, x_{0,p})^\top$	$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}$	$(\vec{x}_0^\top \hat{\vec{\beta}}) \pm t_{1-\frac{\alpha}{2}, n-(p+1)} \times \hat{\sigma}_{\text{OLS}} \sqrt{\vec{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \vec{x}_0}$
Prediction Interval for New Response at x_0 or $\vec{x}_0 = (1, x_{0,1}, \dots, x_{0,p})^\top$	$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{1-\frac{\alpha}{2}, n-2} \times \hat{\sigma}_{\text{OLS}} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{s_{xx}}}$	$(\vec{x}_0^\top \hat{\vec{\beta}}) \pm t_{1-\frac{\alpha}{2}, n-(p+1)} \times \hat{\sigma}_{\text{OLS}} \sqrt{1 + \vec{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \vec{x}_0}$

5.4 Contrast t -Tests

Definition 24 (Contrast t -test). We can generalize the hypothesis test for a *single coefficient*—instead, we're now interested in a *linear combination of coefficients*.

- To test $H_0 : \vec{c}^\top \vec{\beta} = \gamma_0$ vs. $H_A : \vec{c}^\top \vec{\beta} \neq \gamma_0$, we use

$$T = \frac{\vec{c}^\top \hat{\vec{\beta}} - \gamma_0}{\hat{\sigma}_{\text{OLS}} \sqrt{\vec{c}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \vec{c}}} \stackrel{H_0}{\sim} t_{n-(p+1)}$$

as our *test statistic* (i.e., its *exact* distribution is known under H_0).

 **Concept Checker 6.** We know that $\hat{\vec{\beta}} \sim \text{MVN}\left(\vec{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right)$. Let $\Sigma = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. What is the distribution of $\vec{c}^\top \hat{\vec{\beta}}$?

Solution

As a linear combination, $\vec{c}^\top \hat{\vec{\beta}} \sim \mathcal{N}(\mathbb{E}[\vec{c}^\top \hat{\vec{\beta}}], \text{Var}[\vec{c}^\top \hat{\vec{\beta}}])$. Specifically, $\mathbb{E}[\vec{c}^\top \hat{\vec{\beta}}] = \vec{c}^\top \mathbb{E}[\hat{\vec{\beta}}] = \vec{c}^\top \vec{\beta}$ by linearity and $\text{Var}[\vec{c}^\top \hat{\vec{\beta}}] = \vec{c}^\top \Sigma \vec{c}$ by quadratic form (see Problem 2 and Problem 3).

 **Concept Checker 7.** In the hypothesis test for a single coefficient, we see whether the change in expected outcome from a one-unit increase in a single predictor is significant. How can we use the contrast t -test to see whether the change in expected outcome from several predictors changing at once is significant?

Solution

Recall that the model can be written as $\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}$. We're interested in changing $x_{i,1}, \dots, x_{i,p}$ by some amounts $d_1, \dots, d_p$. Let $\vec{c} = (0, d_1, \dots, d_p)^\top$ and $\gamma_0 = 0$. This tests whether the specified simultaneous changes in predictors are associated with a nonzero change in the expected outcome—i.e., $H_0 : d_1\beta_1 + \dots + d_p\beta_p = 0$ vs. $H_A : d_1\beta_1 + \dots + d_p\beta_p \neq 0$.

6 Practice Problems

 **Problem 1.** For this problem, refer to the code and output below.

(a) You're in the final round interview for your dream job, and you're asked to regress RFFT score on age and an indicator for smoking. But alas! You forget the code to produce a confidence interval. How can you impress your interviewers by recovering the 95% confidence interval for β_1 from the output?

(b) Interpret the confidence interval.

(c) How does `Pr(>|t|):Age = <2e-16` provide a sanity check for your confidence interval?

```
```{r}
Read df
df <- read.csv("./prevend.csv")

Run linear regression model
model <- lm(RFFT ~ Age + Smoking, data = df)
summary(model)
```
```

```
Call:
lm(formula = RFFT ~ Age + Smoking, data = df)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-69.878 -15.424  -0.979  14.392  78.060
```

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 134.58824    1.69799  79.263  < 2e-16 ***
Age          -1.18359    0.02994 -39.531  < 2e-16 ***
Smoking      -5.52078    0.79857  -6.913 5.47e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.13 on 4092 degrees of freedom
Multiple R-squared:  0.278,    Adjusted R-squared:  0.2776
F-statistic: 787.6 on 2 and 4092 DF,  p-value: < 2.2e-16

```

Solution

- (a) Generally, for an estimator with an appropriate symmetric sampling distribution around the true parameter, the confidence interval for that parameter is in the form: estimator $\pm$ critical value $\times$ $\widehat{SE}$. With 4092 degrees of freedom, this t -distribution is very well approximated by a Standard Normal, so at $\alpha = 0.05$, we can use $z_{0.975} = 1.96$ as our critical value. Thus, our 95% confidence interval for β_1 is $[-1.18359 - 1.96 \times 0.02994, -1.18359 + 1.96 \times 0.02994] = [-1.2422724, -1.1249076]$. More formally, we know that $T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{OLS} \sqrt{(\mathbf{X}^\top \mathbf{X})_{(j+1,j+1)}^{-1}}} \sim t_{n-(p+1)}$, so a $100(1 - \alpha)\%$ confidence interval for β_1 is in the form: $\hat{\beta}_1 \pm t_{0.975, n-(2+1)} \times \hat{\sigma}_{OLS} \sqrt{(\mathbf{X}^\top \mathbf{X})_{(1+1,1+1)}^{-1}}$. Since $\hat{\beta}_1 \sim \mathcal{N}(\beta_1, \sigma^2(\mathbf{X}^\top \mathbf{X})_{(1+1,1+1)}^{-1})$, its variance is $\text{Var}[\hat{\beta}_1] = \sigma^2(\mathbf{X}^\top \mathbf{X})_{(1+1,1+1)}^{-1}$, so the estimator for its standard error is $\hat{\sigma}_{OLS} \sqrt{(\mathbf{X}^\top \mathbf{X})_{(1+1,1+1)}^{-1}}$.
- (b) We are 95% confident that β_1 is captured in the interval $[-1.2422724, -1.1249076]$, where β_1 is the change in expected RFFT score for each one-unit increase in age, holding smoking status constant.
- (c) The p -value assumes the null hypothesis that $H_0 : \beta_1 = 0$. With such a low p -value, we have evidence to reject H_0 . Our 95% confidence interval is a range of plausible values for β_1 , so it's reassuring that this range does not contain 0.

 **Problem 2.** For an n -dimensional random vector $\vec{X}$, we define

$$\mathbb{E}[\vec{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}, \quad \text{Var}[\vec{X}] = \mathbb{E} \left[(\vec{X} - \mathbb{E}[\vec{X}])(\vec{X} - \mathbb{E}[\vec{X}])^\top \right].$$

- (a) Show that $\mathbb{E}[A\vec{X}] = A\mathbb{E}[\vec{X}]$ for any matrix A and random vector $\vec{X}$.
- (b) Show that $\text{Var}[A\vec{X}] = A\text{Var}[\vec{X}]A^\top$ for any matrix A and random vector $\vec{X}$.
- (c) For any matrix A and vectors $\vec{x}$ and $\vec{y}$ of compatible dimensions, $\vec{x}^\top A\vec{y} = \vec{y}^\top A^\top \vec{x}$. Why?

Solution

(a) Without loss of generality, let $\vec{X}$ be n -dimensional and $A \in \mathbb{R}^{m \times n}$.

$$A\vec{X} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} a_{1,1}X_1 + \cdots + a_{1,n}X_n \\ \vdots \\ a_{m,1}X_1 + \cdots + a_{m,n}X_n \end{pmatrix}.$$

$$A\mathbb{E}[\vec{X}] = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix} = \begin{pmatrix} a_{1,1}\mathbb{E}[X_1] + \cdots + a_{1,n}\mathbb{E}[X_n] \\ \vdots \\ a_{m,1}\mathbb{E}[X_1] + \cdots + a_{m,n}\mathbb{E}[X_n] \end{pmatrix}.$$

Notice that the i th entry of $A\vec{X}$ —i.e., $(A\vec{X})_i$ —is $a_{i,1}X_1 + \cdots + a_{i,n}X_n$ for all $i \in \{1, \dots, m\}$. By linearity, $\mathbb{E}[(A\vec{X})_i] = \sum_{j=1}^n a_{i,j}\mathbb{E}[X_j]$, which is exactly the same as the i th entry of $A\mathbb{E}[\vec{X}]$ —i.e., $(A\mathbb{E}[\vec{X}])_i$. Thus, $\mathbb{E}[A\vec{X}] = A\mathbb{E}[\vec{X}]$.

(b)

$$\begin{aligned} \text{Var}[A\vec{X}] &= \mathbb{E}[(A\vec{X} - \mathbb{E}[A\vec{X}]) (A\vec{X} - \mathbb{E}[A\vec{X}])^\top] && \text{by definition} \\ &= \mathbb{E}[(A\vec{X} - A\mathbb{E}[\vec{X}]) (A\vec{X} - A\mathbb{E}[\vec{X}])^\top] && \text{by linearity} \\ &= \mathbb{E}[A(\vec{X} - \mathbb{E}[\vec{X}]) (A(\vec{X} - \mathbb{E}[\vec{X}]))^\top] && \text{by factoring} \\ &= \mathbb{E}[A(\vec{X} - \mathbb{E}[\vec{X}]) (\vec{X} - \mathbb{E}[\vec{X}])^\top A^\top] && \text{by transpose of a product} \\ &= A\mathbb{E}[(\vec{X} - \mathbb{E}[\vec{X}]) (\vec{X} - \mathbb{E}[\vec{X}])^\top] A^\top && \text{by linearity} \\ &= A\text{Var}[\vec{X}] A^\top && \text{by definition} \end{aligned}$$

(c) With compatible dimensions, $\vec{x} \in \mathbb{R}^{m \times 1}$ and $\vec{y} \in \mathbb{R}^{n \times 1}$. Note that $\vec{x}^\top A \vec{y}$ is a scalar since $\vec{x}^\top A \vec{y} = (\vec{x}^\top A) \vec{y}$ by associativity, and $\vec{x}^\top A \in \mathbb{R}^{1 \times n}$, so $(\vec{x}^\top A) \vec{y} \in \mathbb{R}$. A scalar is always equal to its transpose.

$$\begin{aligned} \vec{x}^\top A \vec{y} &= (\vec{x}^\top A \vec{y})^\top && \text{by transpose of a scalar} \\ &= \vec{y}^\top A^\top (\vec{x}^\top)^\top && \text{by transpose of a product} \\ &= \vec{y}^\top A^\top \vec{x} && \text{by transpose of a transpose} \end{aligned}$$

 **Problem 3.** Put simply, a *quadratic form* is an expression in the form $\vec{x}^\top A \vec{x}$, where $\vec{x}$ is a vector and A is a square matrix. Let $\vec{X}$ be a random vector with $\text{Var}[\vec{X}] = \Sigma$. Then, for any vector $\vec{a}$,

$$\text{Var}[\vec{a}^\top \vec{X}] = \vec{a}^\top \Sigma \vec{a},$$

which is a quadratic form!

(a) A covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$ must be positive semidefinite, meaning that

$$\vec{a}^\top \Sigma \vec{a} \geq 0$$

for all $\vec{a} \in \mathbb{R}^n$. Why?

(b) For simplicity, let $\vec{X} = (X_1, X_2)^\top$ and $\vec{a} = (a_1, a_2)^\top$. Find $\text{Var}[a_1X_1 + a_2X_2]$ first by using single-variable probability and then by using linear algebra.

Solution

(a) Recall that variance can never be negative, so $\vec{a}^\top \Sigma \vec{a} = \text{Var}[\vec{a}^\top \vec{X}] \geq 0$. Intuitively, $\vec{a}^\top \vec{X}$ corresponds to an arbitrary linear combination of the random variables, and $\vec{a}^\top \Sigma \vec{a}$ measures the variance of that linear combination.

(b) First, let's use single-variable probability:

$$\text{Var}[a_1X_1 + a_2X_2] = a_1^2\text{Var}[X_1] + a_2^2\text{Var}[X_2] + 2a_1a_2\text{Cov}[X_1, X_2] \quad \text{by bilinearity}$$

Next, let's use linear algebra:

$$\begin{aligned} \text{Var}[a_1X_1 + a_2X_2] &= \text{Var}[\vec{a}^\top \vec{X}] && \text{by rewriting} \\ &= \vec{a}^\top \Sigma \vec{a} && \text{by quadratic form} \\ &= (a_1 \ a_2) \begin{pmatrix} \text{Var}[X_1] & \text{Cov}[X_1, X_2] \\ \text{Cov}[X_1, X_2] & \text{Var}[X_2] \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} && \text{by substituting} \\ &= a_1^2\text{Var}[X_1] + a_2^2\text{Var}[X_2] + 2a_1a_2\text{Cov}[X_1, X_2] && \text{by matrix algebra} \end{aligned}$$

A Proofs

A.1 Geometric Derivation of OLS Estimators

Proof. We want to show that $\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{Y}$. We know that $\mathbf{X}^\top \vec{e} = 0$.

$$\begin{aligned} \mathbf{X}^\top \vec{e} = 0 &\implies \mathbf{X}^\top (\vec{Y} - \hat{\vec{Y}}) = 0 && \text{since } \vec{e} = \vec{Y} - \hat{\vec{Y}} \\ &\implies \mathbf{X}^\top (\vec{Y} - \mathbf{X} \hat{\vec{\beta}}) = 0 && \text{since } \hat{\vec{Y}} = \mathbf{X} \hat{\vec{\beta}} \\ &\implies \mathbf{X}^\top \vec{Y} - \mathbf{X}^\top \mathbf{X} \hat{\vec{\beta}} = 0 && \text{by distributing} \\ &\implies \mathbf{X}^\top \mathbf{X} \hat{\vec{\beta}} = \mathbf{X}^\top \vec{Y} && \text{by adding } \mathbf{X}^\top \mathbf{X} \hat{\vec{\beta}} \\ &\implies \hat{\vec{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \vec{Y} && \text{by multiplying by } (\mathbf{X}^\top \mathbf{X})^{-1} \end{aligned}$$

□